Year 12 Further Maths - Vectors

Similar questions - Intersecting lines, common perpendiculars and distances

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The lines l_1 and l_2 have equations $l_1:\ \mathbf{r} = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$ and $l_2:\ \mathbf{r} = \begin{pmatrix} 7 \\ 1 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} -3 \\ 2 \\ -4 \end{pmatrix}$.

(a) Show that l_1 and l_2 intersect. **[4]**

(b) Write down the point of intersection. **[1]**

2 The lines L_1 and L_2 have equations $L_1:\ \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$ and $L_2:\ \mathbf{r} = \begin{pmatrix} 2 \\ 7 \\ -8 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$.

(a) Show that L_1 and L_2 intersect, giving the position vector of the point of intersection. **[5]**

(b) Find the equation of the line which intersects L_1 and L_2 and is perpendicular to both. Give your answer in cartesian form. **[3]**

3 The line l_1 passes through $A(1,2,-3)$ and $B(4,0,3)$.

(a) Find a vector equation for l_1. **[2]**

(b) Determine whether the point $(13,-6,21)$ lies on l_1. **[2]**

The line l_2 passes through the origin O and intersects l_1 at the point C. The lines l_1 and l_2 are perpendicular.

(c) By using the fact that C lies on l_1, find a vector equation for l_2. **[4]**

(d) Hence find the shortest distance from O to l_1. **[2]**

4 The position vector of point A is $\mathbf{a} = -3\mathbf{i} + \mathbf{j} + 2\mathbf{k}$. The line l passes through A and is perpendicular to $\mathbf{a}$. The vector $\mathbf{b} = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$ is also perpendicular to l.

(a) Determine the shortest distance between the origin and l. **[2]**

(b) Find a vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$. **[2]**

(c) Write down an equation of l in vector form. **[1]**

(d) P is a point on l such that $PA = 2OA$. Find the angle POA to 3 s.f. **[3]**

(e) The point C has position vector $\mathbf{c} = p\mathbf{a}$. The line m through C has equation $\mathbf{r} = \mathbf{c} + \mu\mathbf{b}$, and the point with position vector $-3\mathbf{i} - 4\mathbf{j} + 7\mathbf{k}$ lies on m. Find p. **[4]**

Question 1

(a) Equate: $2+\lambda = 7-3\mu$, $-1+2\lambda = 1+2\mu$, $4-3\lambda = 2-4\mu$. From the first two: $\lambda = 5-3\mu$ and $-1+10-6\mu = 1+2\mu \Rightarrow \mu = 1$, $\lambda = 2$. Check the third: $4-6=-2$ and $2-4=-2$ (correct), so the lines intersect

(b) $(4,3,-2)$

Question 2

(a) $1+2\lambda = 2+\mu$, $2+\lambda = 7-\mu$, $3-\lambda = -8+3\mu$. Adding the first two: $3+3\lambda = 9$, $\lambda = 2$, $\mu = 3$. Third: $3-2=1$ and $-8+9=1$ (correct). Point $\begin{pmatrix}5\\4\\1\end{pmatrix}$

(b) Direction $\begin{pmatrix}2\\1\\-1\end{pmatrix} \times \begin{pmatrix}1\\-1\\3\end{pmatrix} = \begin{pmatrix}2\\-7\\-3\end{pmatrix}$. Line: $\dfrac{x-5}{2} = \dfrac{y-4}{-7} = \dfrac{z-1}{-3}$

Question 3

(a) $\overrightarrow{AB} = \begin{pmatrix}3\\-2\\6\end{pmatrix}$; $\mathbf{r} = \begin{pmatrix}1\\2\\-3\end{pmatrix} + \lambda \begin{pmatrix}3\\-2\\6\end{pmatrix}$

(b) $1+3\lambda = 13 \Rightarrow \lambda = 4$: $y = 2-8 = -6$ (correct), $z = -3+24 = 21$ (correct). Yes, it lies on l_1

(c) $\overrightarrow{OC} = \begin{pmatrix}1+3\lambda\\2-2\lambda\\-3+6\lambda\end{pmatrix}$ perpendicular to $\begin{pmatrix}3\\-2\\6\end{pmatrix}$: $3(1+3\lambda)-2(2-2\lambda)+6(-3+6\lambda) = 49\lambda - 19 = 0$, $\lambda = \frac{19}{49}$.

$C = \frac{1}{49}(106,60,-33)$; $l_2:\ \mathbf{r} = \mu \begin{pmatrix}106\\60\\-33\end{pmatrix}$

(d) $|OC| = \dfrac{\sqrt{106^2+60^2+33^2}}{49} = \dfrac{\sqrt{15925}}{49} = 2.58$

Question 4

(a) l is perpendicular to $\mathbf{a}$ so OA is the shortest distance: $|\mathbf{a}| = \sqrt{14} = 3.74$

(b) $\mathbf{a} \times \mathbf{b} = \begin{pmatrix}5\\5\\5\end{pmatrix}$, so $\begin{pmatrix}1\\1\\1\end{pmatrix}$ is suitable

(c) $\mathbf{r} = \begin{pmatrix}-3\\1\\2\end{pmatrix} + t \begin{pmatrix}1\\1\\1\end{pmatrix}$

(d) Triangle OAP has a right angle at A: $\tan POA = \dfrac{PA}{OA} = 2$, $POA = 63.4^\circ$

(e) $-3p+\mu = -3$, $p-2\mu = -4$, $2p+\mu = 7$. From the first and third: $5p = 10$, $p = 2$ (then $\mu = 3$; check second: $2-6=-4$ (correct))